Kazakh-British Technical University

Data collection and preparation

STUDENTS' INDEPENDENT STUDY (SIS) №1

Music analytics - exploring the relationship between artist popularity and career length/number of albums

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1. **The goal and research question** of our SIS is to know if an artist's popularity on Last.fm related to their career length and number of studio albums?

2. **Data and collection.**
   API: https://www.last.fm/api/show/artist.getInfo
   Web scraping: Wikipedia via *requests + BeautifulSoup.* "Years active" comes from the artist page infobox, and "Studio albums" comes from the discography page infobox.

3. **Data cleaning and transformation.**
   1) Deleting duplicates, column remaining, 0 missing values and changed typed of data to numeric.
   2) Added customer function *calculate_career_years* parses intervals like "1974–1986 1988–1989 1992–present" and sums the years. It raises an error if the format looks wrong.
   3) New features:
      a) listeners_m: listeners in millions.
      b) intensity: playcount / listeners = how many times the average listener played the artist.
      c) Career groups (1–15, 15–30, 30+) and album groups (1–5, 6–10, 10+).
   4) Merge: inner merge on artist name, giving 50 rows.

4. **Results**

Relationship	Number
Career length vs listeners	$r \approx$ **0.10**, almost no relationship.
Albums vs listeners	$r \approx$ **0.09**, also almost none.
Career length vs albums	$r \approx$ **0.67**, a strong and logical relationship.
Career length vs intensity	$r \approx -0.32$. Artists with shorter careers get replayed more
Median intensity by career group	1–15 years = **96.6**, 15–30 years=**73.1**, 30+ years=**32.6**
Median listeners peak in the middle group	15–30 years (4.86M) and 6–10 albums (4.86M).

5. **Conclusions and limitations**

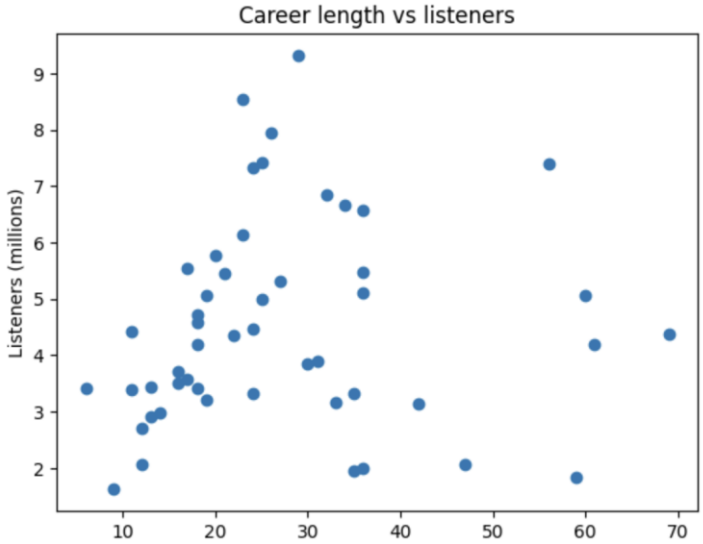
A longer career does not mean more listeners.

2) Newer artists are listened to more intensively. Older artists have a broader, more casual audience.

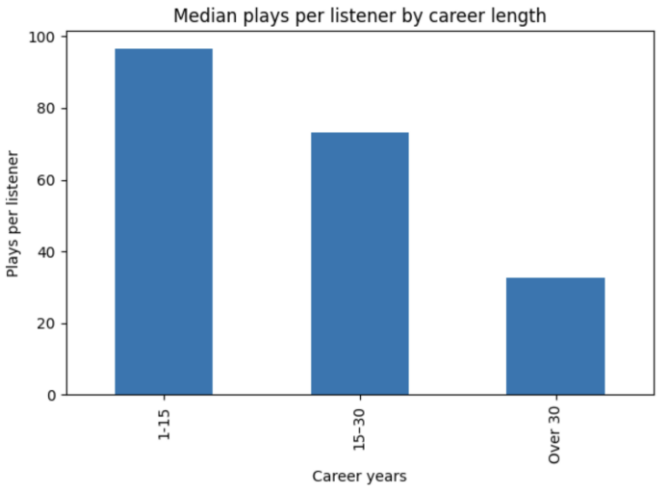
3) Limitations:

The sample is only 50 well-known artists, chosen by hand.
Last.fm is not the whole market.
There are outliers: Taylor Swift (intensity 620) and Frank Sinatra (74 albums).
Wikipedia's "Years active" sometimes includes childhood careers, for example, Bruno Mars and Beyoncé from 1990.

6. Two the most important visualisations



Career length has almost no impact on the number of listeners.



Young artists are listened to more intensely.